

8. This question is about momentum.

(a) State the principle of conservation of momentum.

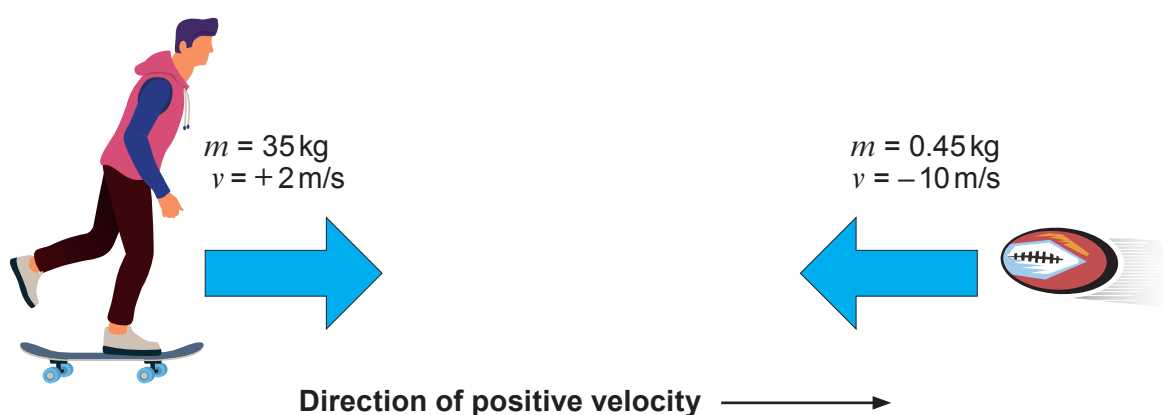
[2]

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(b) A skateboarder of mass 35 kg moving with a velocity of 2 m/s catches and holds on to a 0.45 kg rugby ball thrown directly towards him at 10 m/s.



Use the equation:

$$\text{momentum} = \text{mass} \times \text{velocity}$$

to calculate the **total** momentum of the skateboarder and the ball before the collision and give the unit.

[4]

Total momentum =

Unit =

